

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

The logs show that initially, the attacker's SYN request is answered normally by the web server (log items 52-54). An employee visitor with the IP address of 198.51.100.14 successfully completes a SYN/ACK connection handshake with the webserver (log item nos. 55, 56, 58). Then, the employee's browser requests the sales.html webpage with the GET command and the web server responds (log item no. 60 and 62).

In the next 20 rows, the log begins to reflect the struggle the web server is having to keep up with the abnormal number of SYN requests coming in at a rapid pace. The attacker is sending several SYN requests every second. Communications between legitimate employee website visitors and the web server is failing. An HTTP/1.1 504 Gateway Timeout (text/html) error message and an [RST, ACK] packet was received by the employees trying to visit the webpage. After log 125 the web server stops responding to legitimate visitors due to the overwhelming amount of SYN packets sent to the server by the attacker. This network interruption is caused by a direct DoS SYN flood attack from a single IP address (203.0.113.0).

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The [SYN] packet is the initial request from an employee visitor trying to connect to a web page hosted on the web server. SYN stands for "synchronize." The [SYN, ACK] packet is the web server's response to the visitor's request agreeing to the connection. The server will reserve system resources for the final step of the handshake. SYN, ACK stands for "synchronize acknowledge." The [ACK] packet is the visitor's machine acknowledging the permission to connect. This is the final step required to make a successful TCP connection. ACK stands for "acknowledge." As the attacker keeps sending several SYN requests every second, communications between legitimate employee website visitors and the web server is failing. An HTTP/1.1 504 Gateway Time-out (text/html) error message and an [RST, ACK] packet is received by visitors. An HTTP/1.1 504 Gateway Time-out (text/html) error message is generated by a gateway server that is waiting for a response from the web server. If the web server takes too long to respond, the gateway server sends a timeout error message to the requesting browser. An [RST, ACK] packet is sent to the requesting visitor if the [SYN, ACK] packet is not received by the web server. RST stands for reset, acknowledge. The visitor receives a timeout error message in their browser and the connection attempt is dropped. The visitor can refresh their browser to attempt to send a new SYN request. According to the logs, after log 125 the web server stops responding to legitimate visitors due to the overwhelming amount of SYN packets sent to the server by the attacker. This network interruption is caused by a direct DoS SYN flood attack from a single IP address (203.0.113.0).